

Eric J. Verbeke

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🌐 <https://ericverbeke.github.io/>

Research Positions

- 2025 - present 📖 **The University of Osaka**, CZ Biohub and SS-F Global Science Scholar
Premium Research Institute for Human Metaverse Medicine
- 2021 - 2025 📖 **Princeton University**, Postdoctoral Research Associate
Program in Applied and Computational Mathematics
Center for Statistics and Machine Learning
- summer 2024 📖 **The University of Tokyo**, JSPS Fellow
Graduate School of Medicine
- 2015-2016 📖 **National Institute for Standards and Technology**, Guest Researcher
Molecular and BioPhotonics group

Education

- 2016 – 2021 📖 **Ph.D., University of Texas at Austin**, Biochemistry
- 2011 – 2015 📖 **B.S., University of Colorado at Boulder**, Chemical and Biological Engineering

Publications

🌐 [Link to my Google Scholar page](#)

- 1 Y. Shi, A. Singer, and **E. J. Verbeke**, “Fast rigid alignment of heterogeneous images in sliced wasserstein distance,” *SIAM Journal on Imaging Sciences*, vol. 19, no. 3, pp. 1469–1515, 2026. 📄 DOI: 10.1137/25M1742916.
- 2 T. Muller, A. L. Duncan, **E. J. Verbeke**, and J. Kileel, “Algebraic constraints and algorithms for common lines in cryo-EM,” *Biological Imaging*, vol. 4, e9, 2024, ISSN: 2633-903X. 📄 DOI: 10.1017/S2633903X24000072.
- 3 **E. J. Verbeke**, M. A. Gilles, T. Bendory, and A. Singer, “Self Fourier shell correlation: properties and application to cryo-ET,” *Communications Biology*, vol. 7, no. 1, p. 101, Jan. 2024, ISSN: 2399-3642. 📄 DOI: 10.1038/s42003-023-05724-y.
- 4 A. Zhang, O. Mickelin, J. Kileel, *et al.*, “Moment-based metrics for molecules computable from cryogenic electron microscopy images,” *Biological Imaging*, vol. 4, e3, Feb. 2024, ISSN: 2633-903X. 📄 DOI: 10.1017/S2633903X24000023.
- 5 M. J. Lucas, H. S. Pan, **E. J. Verbeke**, *et al.*, “Cross-seeding controls A β fibril populations and resulting functions,” *The Journal of Physical Chemistry B*, vol. 126, no. 11, pp. 2217–2229, 2022. 📄 DOI: 10.1021/acs.jpcc.1c09995.
- 6 W. Sae-Lee, C. L. McCafferty, **E. J. Verbeke**, *et al.*, “The protein organization of a red blood cell,” *Cell Reports*, vol. 40, no. 3, p. 111 103, 2022. 📄 DOI: 10.1016/j.celrep.2022.111103.
- 7 M. J. Lucas, H. S. Pan, **E. J. Verbeke**, L. J. Webb, D. W. Taylor, and B. K. Keitz, “Functionalized mesoporous silicas direct structural polymorphism of amyloid β fibrils,” *Langmuir*, vol. 36, no. 26, pp. 7345–7355, 2020. 📄 DOI: 10.1021/acs.langmuir.0c00827.
- 8 C. L. McCafferty, **E. J. Verbeke**, E. M. Marcotte, and D. W. Taylor, “Structural biology in the multi-omics era,” *Journal of Chemical Information and Modeling*, vol. 60, no. 5, pp. 2424–2429, May 2020. 📄 DOI: 10.1021/acs.jcim.9b01164.

- 9 **E. J. Verbeke**, Y. Zhou, A. P. Horton, A. L. Mallam, D. W. Taylor, and E. M. Marcotte, “Separating distinct structures of multiple macromolecular assemblies from cryo-EM projections,” *Journal of Structural Biology*, vol. 209, no. 1, p. 107416, 2020.  DOI: 10.1016/j.jsb.2019.107416.
- 10 X. Yi, **E. J. Verbeke**, Y. Chang, D. J. Dickinson, and D. W. Taylor, “Electron microscopy snapshots of single particles from single cells,” *Journal of Biological Chemistry*, vol. 294, no. 5, pp. 1602–1608, 2019.  DOI: 10.1074/jbc.RA118.006686.
- 11 **E. J. Verbeke**, A. L. Mallam, K. Drew, E. M. Marcotte, and D. W. Taylor, “Classification of single particles from human cell extract reveals distinct structures,” *Cell Reports*, vol. 24, no. 1, 259–268.e3, 2018.  DOI: 10.1016/j.celrep.2018.06.022.
- 12 K. P. Wall, M. Pagratis, G. Armstrong, *et al.*, “Molecular determinants of tubulin’s c-terminal tail conformational ensemble,” *ACS Chemical Biology*, vol. 11, no. 11, pp. 2981–2990, 2016.  DOI: 10.1021/acscchembio.6b00507.

Teaching Experience

Princeton University (Lecturer)

Spring 2025  SML 301 - Data Intelligence: Modern Data Science Methods (co-instructor)
 Fall 2024  SML 310 - Research Projects in Data Science (co-instructor)

University of Texas at Austin (Teaching Assistant)

Fall 2018  BCH 370 - Physical Methods of Biochemistry
 Fall 2017  BCH 370 - Physical Methods of Biochemistry

Invited Talks

Jun 2025 - IPAM Conference on Computational Microscopy, Lake Arrowhead, CA
Fast Alignment of Heterogeneous Images in Sliced Wasserstein Distance
 Jun 2024 - IPAM Conference on Computational Microscopy, Lake Arrowhead, CA
Fast Alignment of Images Using Optimal Transport
 Nov 2022 - IPAM Conference on Computational Microscopy, UCLA
Cryo-Electron Microscopy and Beyond
 Jun 2022 - Gordon Research Conference for Three Dimensional Electron Microscopy, Castelldefels, Spain
Don't Purify in Vein! Structural Insights from Red Blood Cell Lysate
 Jun 2019 - Gordon Research Conference for Three Dimensional Electron Microscopy, Hong Kong, China
Separating distinct structures of multiple macromolecular assemblies from cryo-EM projections
 Sep 2017 - UT Austin ICMB Annual Conference, Horseshoe Bay, TX
Visualizing the Human Complexome

Awards

2025-2027  The Chan Zuckerberg Biohub Network and the Stellar Science Foundation
 Global Science Scholars Fellowship
 2024-2025  Princeton University DataX Postdoctoral Fellowship
 2024  Japan Society for Promotion of Science Summer Fellowship
 2020-2021  University Graduate Continuing Fellowship, University of Texas at Austin
 2019  Gordon Research Conference - Travel Award
 2018  David Bruton Jr. Graduate School Fellowship, University of Texas at Austin

Awards (continued)

- 2014  Undergraduate Research Opportunity Program, University of Colorado at Boulder
- 2013  HHMI Undergraduate Research Fellowship, University of Colorado at Boulder

Professional Service

Reviewer

- ANR AAPG, April 2025
- Structure, Cell Press, July 2022